

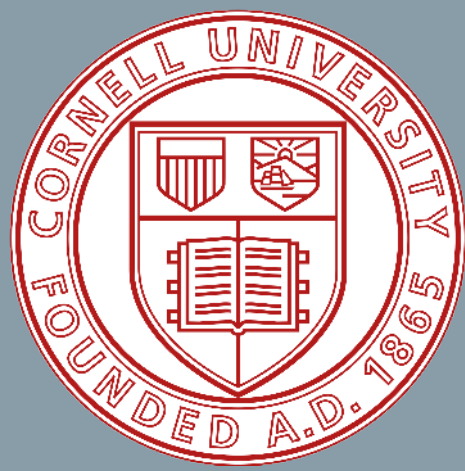
Engineering Drought Resilience Through Plant Architecture

Jua’nele Webb, Magdalena M. Julkowska
contact: juanelewebb@gmail.com; mmj55@cornell.edu



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Introduction

Drought is a major constraint on agricultural productivity, reducing crop yield and limiting sustainable food production. Improving drought resilience requires a better understanding of how plant architecture influences water-use efficiency (WUE) and physiological responses to water limitation. The drought-adapted tepary bean (*Phaseolus acutifolius*) provides an opportunity to identify architectural traits associated with improved drought performance, while soybean (*Glycine max*) serves as a complementary model for evaluating architectural modifications. This study investigated developmental differences in stomatal conductance in tepary bean, hormone-mediated architectural regulation in tepary bean, and the effects of shoot apical meristem removal in soybean under well-watered and drought conditions. **We hypothesize that compact plant architectures and preferential maintenance of water allocation to developing tissues improve drought resilience and water-use efficiency under water-limited conditions.**

Materials & Methods

Plant Materials and Architecture Modifiers

Three independent drought experiments were conducted using tepary bean (*Phaseolus acutifolius*) and soybean (*Glycine max*).

Experiment	Species	Description	Number of Plants
1 Developmental Water Allocation	Tepary Bean (<i>Phaseolus acutifolius</i>)	Comparison of highest vs. lowest mature trifoliate leaves	4 (2 well-watered, 2 drought)
2 Hormone-Mediated Architecture	Tepary Bean (<i>Phaseolus acutifolius</i>)	Hormone treatments: PAC, IAA, and Control	15 (PAC, IAA, and Control groups)
3 Shoot Tip Decapitation	Soybean (<i>Glycine max</i>)	Intact vs. decapitated shoot tip (apical meristem)	14 (7 intact, 7 decapitated)

Architecture Modifiers

Hormone Treatments

PAC: Paclobutrazol
IAA: Indole-3-acetic acid
Control: Mock treatment

Shoot Tip Decapitation

Removal of the shoot tip (apical meristem)

Stomatal Conductance Evaluation

Stomatal conductance was measured using an SC-1 Leaf Porometer to quantify leaf gas exchange. Measurements were collected from the highest and lowest mature leaves, placed on the center trifoliate, to evaluate physiological water allocation across the shoot developmental axis.

Drought Stress Application

Plants were maintained under well watered or drought conditions. Drought stress was imposed by withholding irrigation for three days before physiological and architectural measurements were collected.

Water Treatment: Control (blue drop), Drought (white drop)

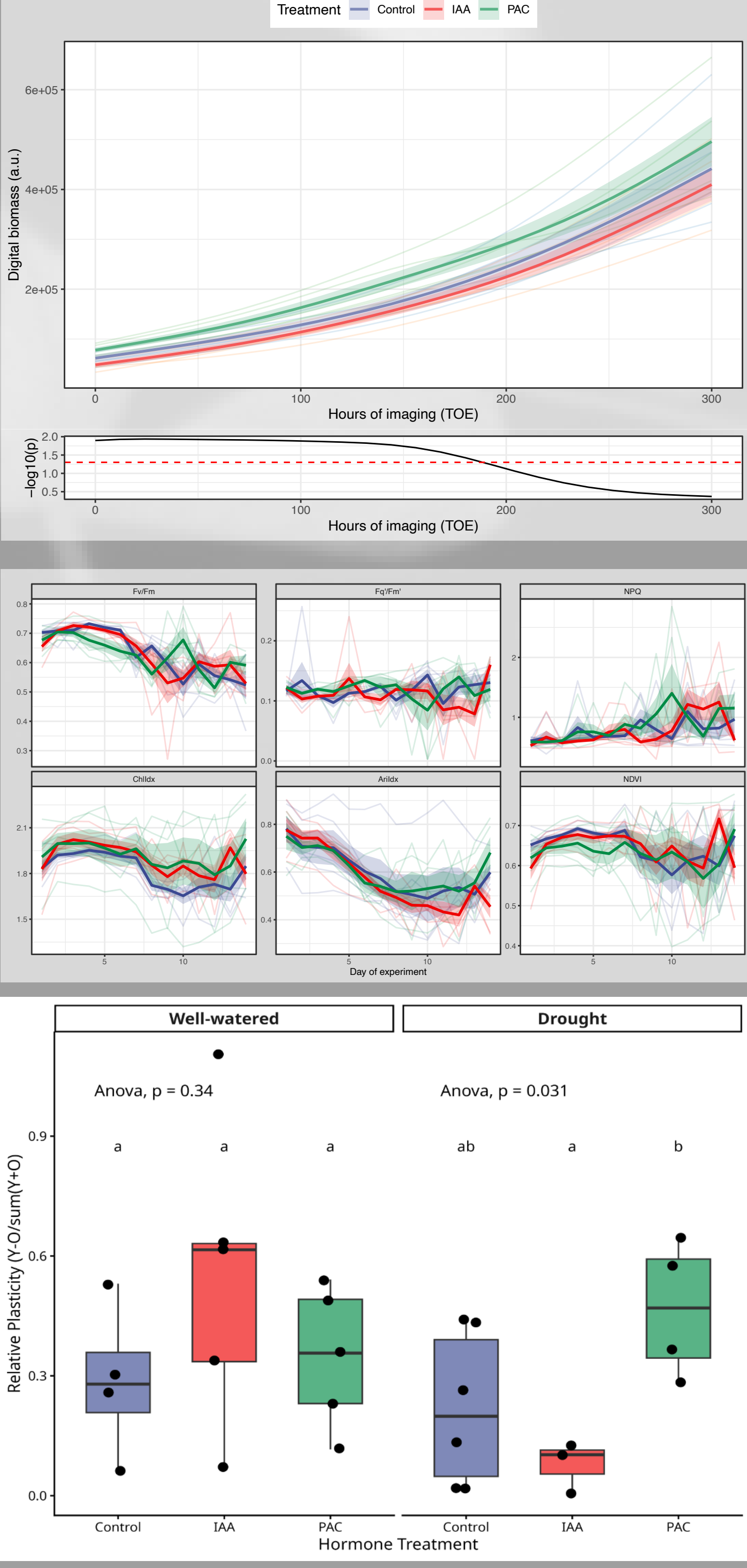
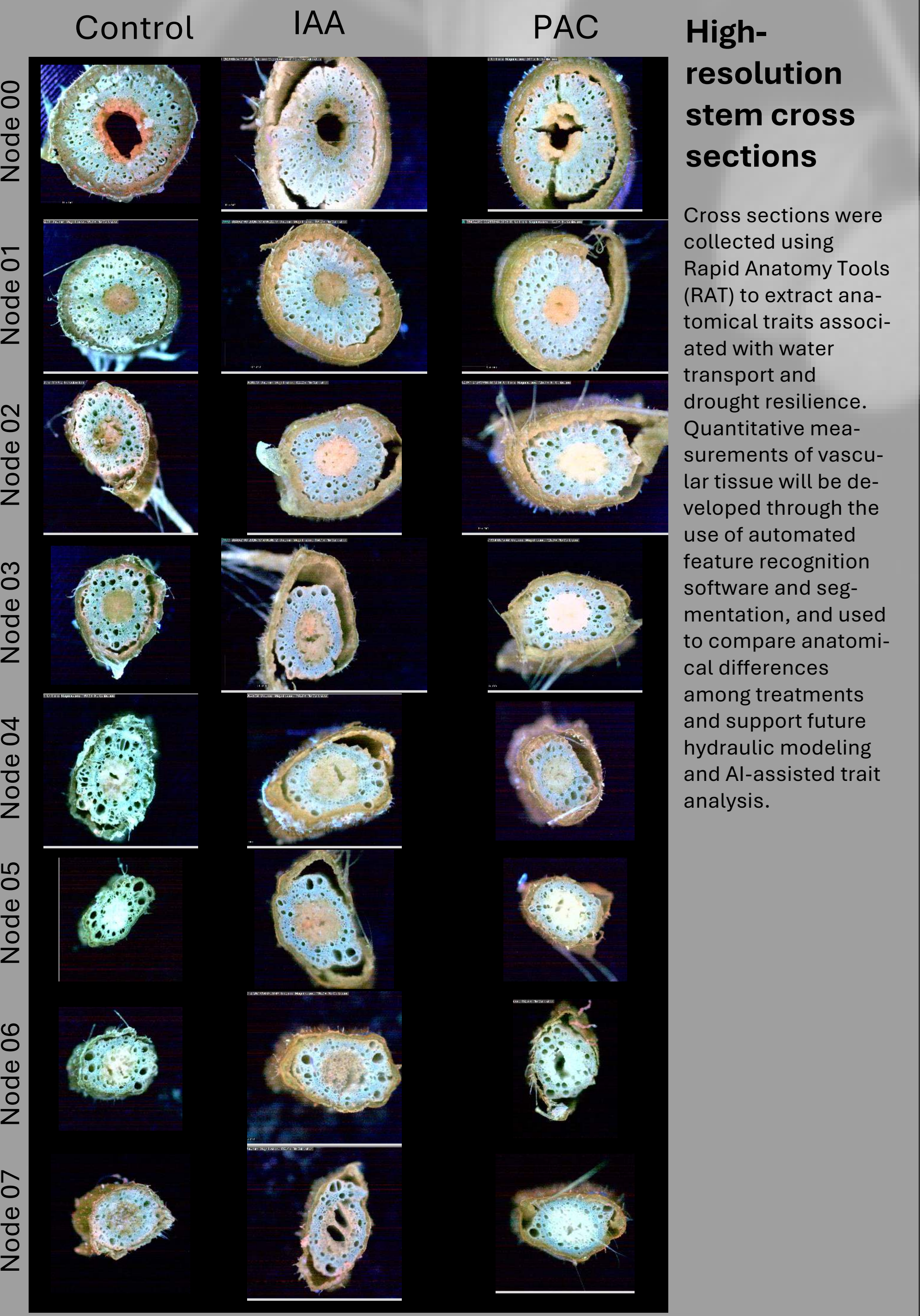
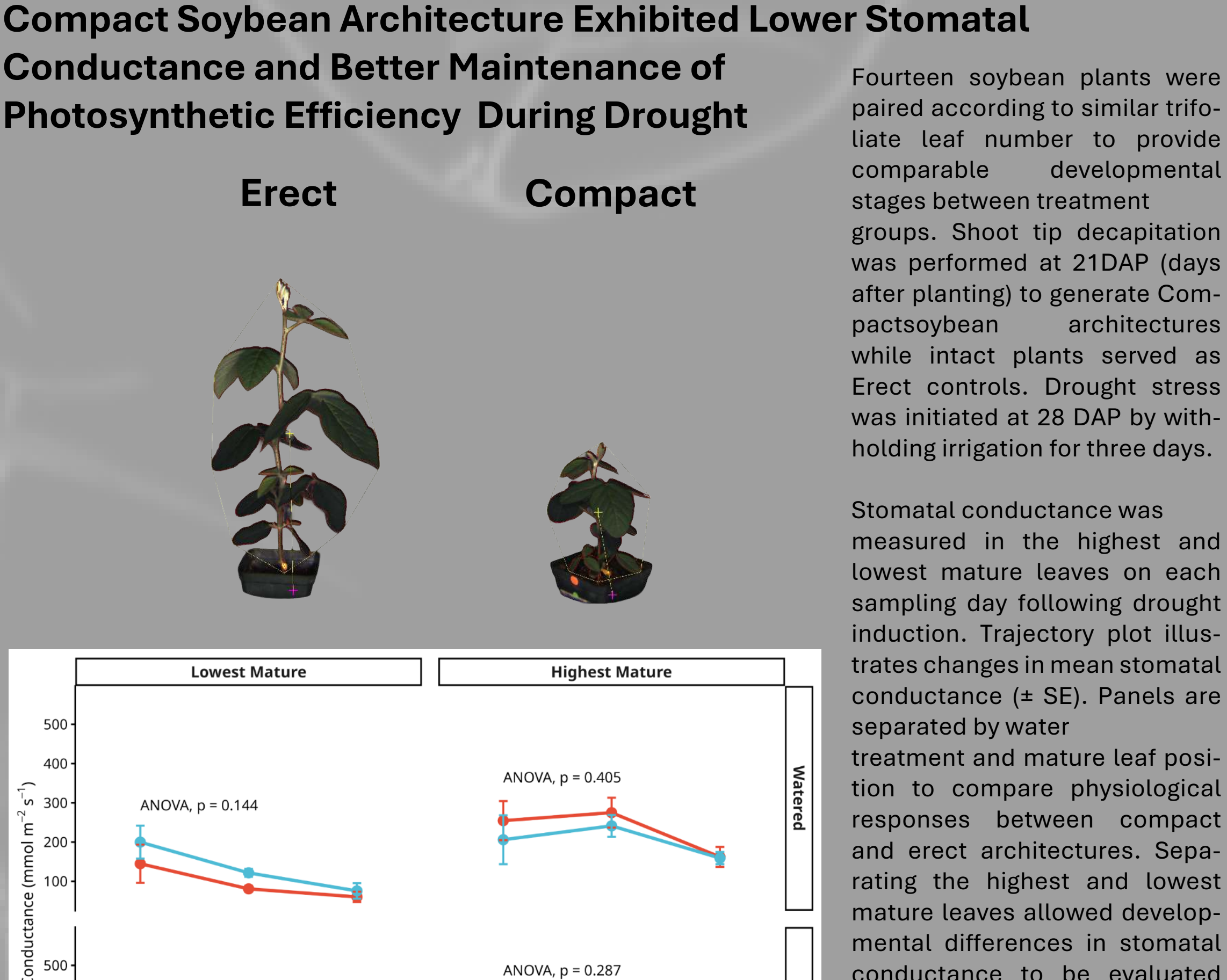
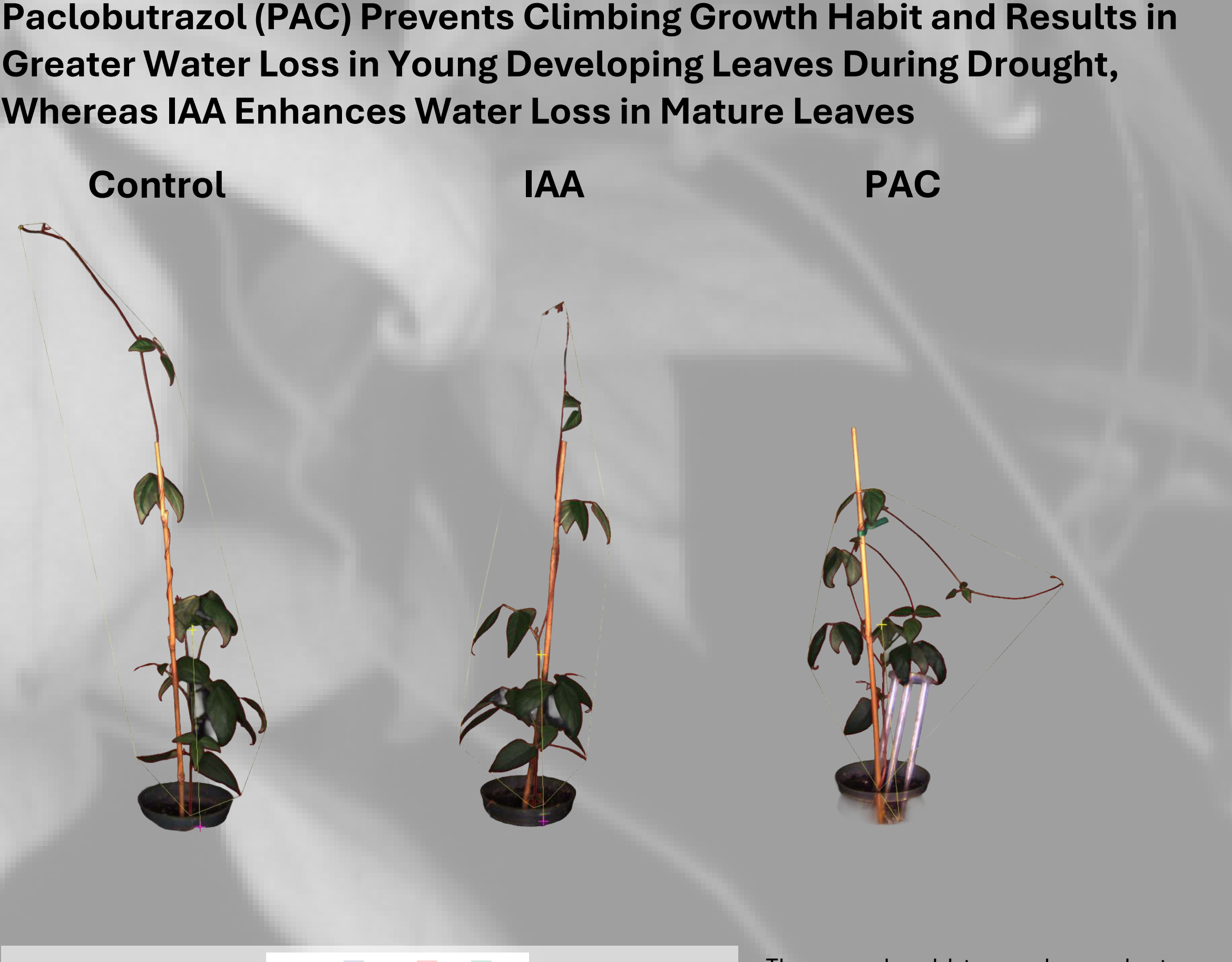
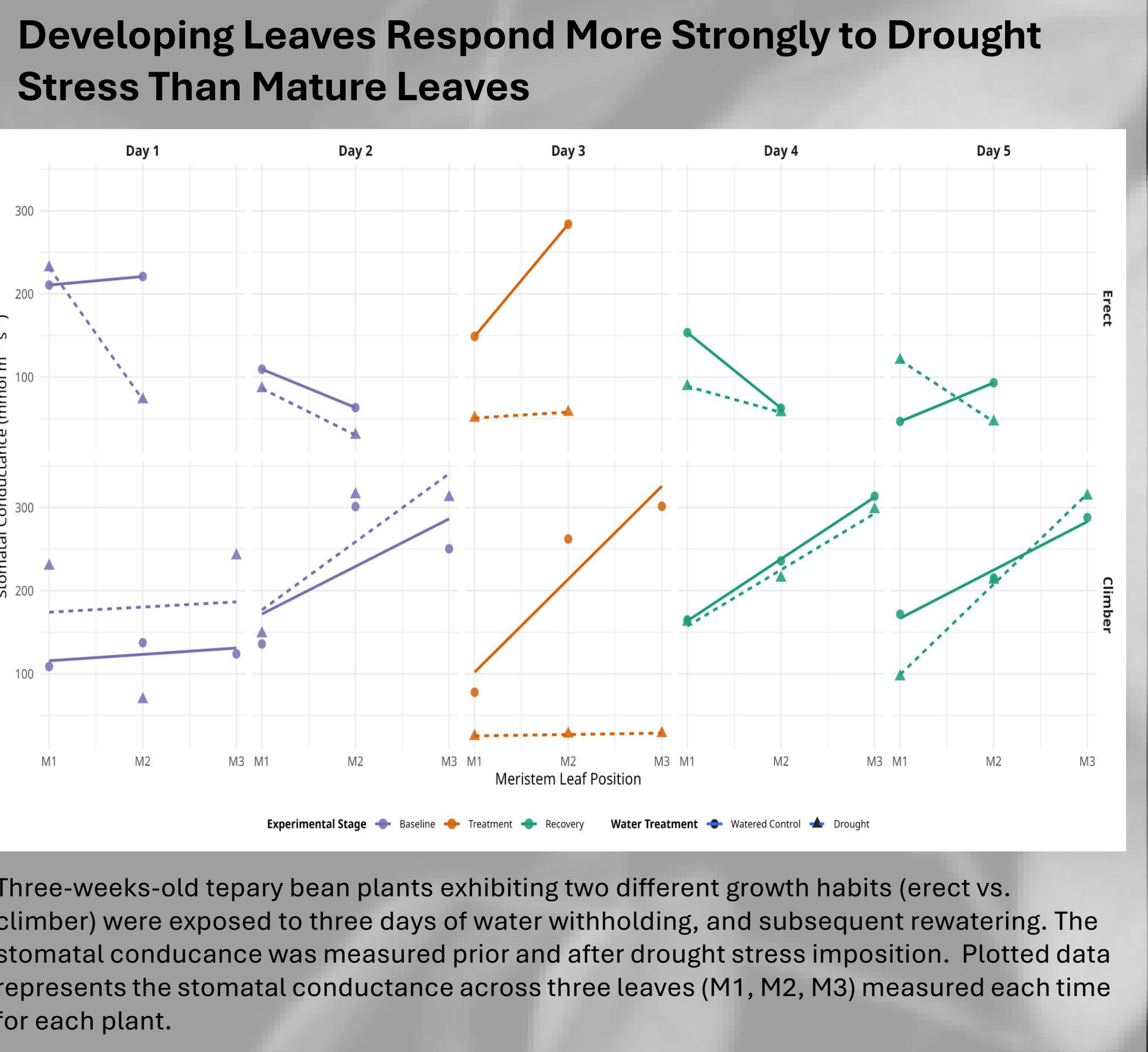
Digital Phenotyping

High-throughput imaging was performed using the PhenoSight platform to quantify plant architecture and photosynthetic performance. RGB images, chlorophyll fluorescence, and Photosystem II (PSII) efficiency were collected throughout the hormone-treated tepary and soybean experiments.

Rapid Anatomy Tools (RAT)

Stem cross sections were imaged using a Dino-Lite Edge AM8517MT-FUW digital microscope to obtain high-resolution anatomical images for analysis with RAT. The imaging system captures vascular tissue, xylem, phloem, and stem structural features that can be quantitatively measured to compare anatomical differences among plant architectures and drought treatments.

Results

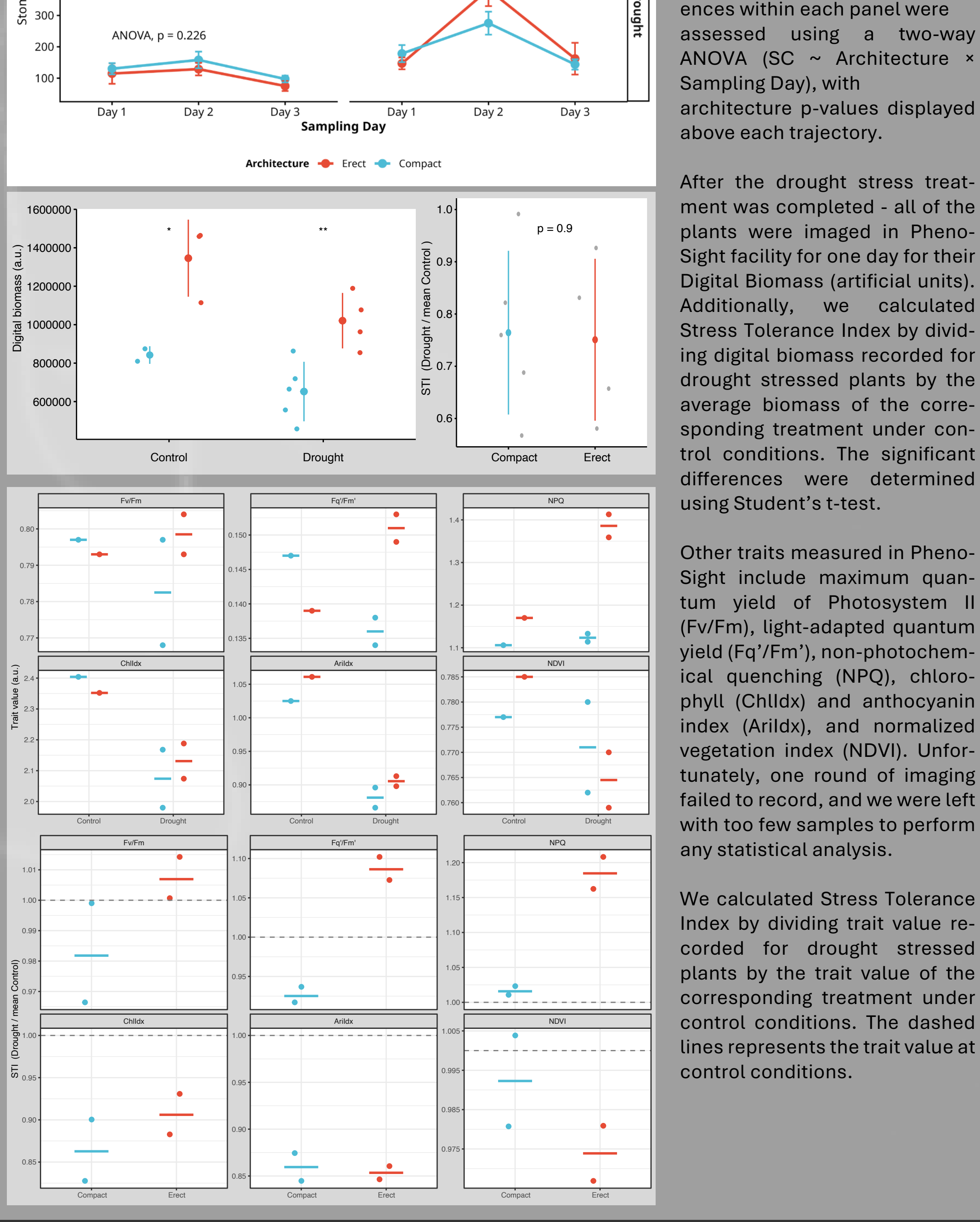


Three-weeks-old tepary bean plants were treated with auxin (IAA) and gibberellin synthesis inhibitor (paclobutrazol – PAC) by spraying them. The plants were measured in PhenoSight facility for hormone-induced changes in plant photosynthesis and growth rate.

The traits recorded in PhenoSight include Digital Biomass (artificial units) calculated as sum of projected shoot area over 6 side view images and one top view image. The progression represents plant growth over time of experiment (TOE). The significant differences between treatments are represented as -log10(p-value) where line above threshold represents significant differences.

Other traits measured in PhenoSight include maximum quantum yield of Photosystem II (Fv/Fm), light-adapted quantum yield (Fq'/Fm'), non-photochemical quenching (NPQ), chlorophyll (Chl) and anthocyanin index (Arldx), and normalized vegetation index (NDVI).

Approximately 2 weeks after hormonal treatment, plants were exposed to drought stress. The stomatal conductance was measured from the youngest and oldest leaves. We have calculated relative plasticity of stomatal conductance by calculating the difference between youngest and oldest leaves, and normalizing it over the total stomatal conductance measured over youngest and oldest leaves. The statistical analysis for the effect of treatment within Control and Drought stress treatment groups was performed using one-way ANOVA, whereas significantly different groups (p-value < 0.05) were determined using Tukey HSD test.



Future Perspectives

- Integrate CPlantBox modeling to simulate radial and axial water transport across contrasting plant architectures.
- Quantify stem vascular anatomy using Rapid Anatomy Tools (RAT) to identify anatomical traits associated with drought resilience.
- Develop AI-assisted image analysis to automate anatomical feature extraction and accelerate hydraulic trait characterization.

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